3.2 PN Junction: I-V Characteristic

3.2.1 Qualitative Derivation

3.2.2 Quantitative Derivation

- Minority Carrier Distribution
- Minority Carrier Currents
- I-V Characteristic
- Saturation Current

3.2.3 Junction Breakdown

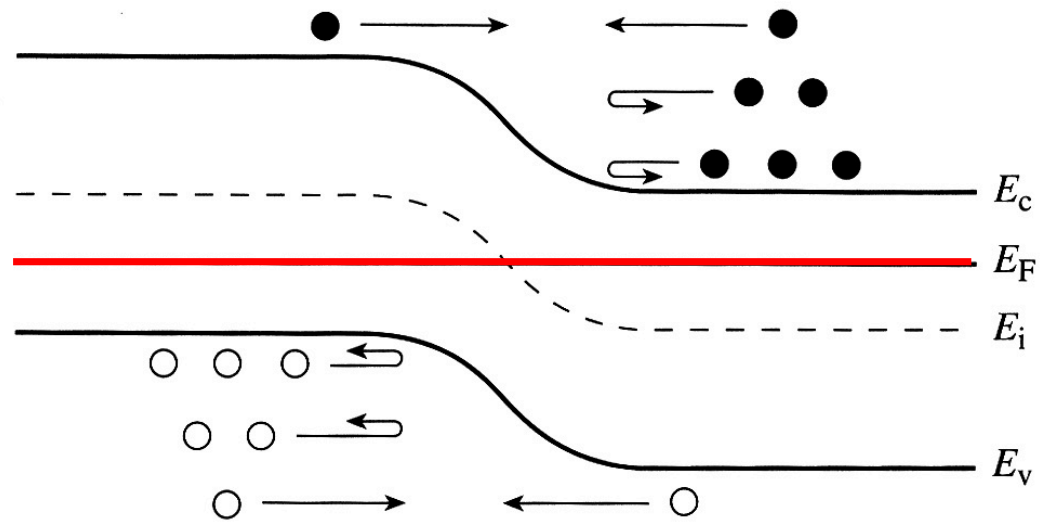
- Avalanche Breakdown
- Zener Breakdown

Pierret, Chapter 6.1, page 235-259

3.2.1 Qualitative Derivation
(a) Equilibrium

- Under equilibrium, the **net current flow is zero**, i.e. the electron/hole drift and diffusion currents balance each other

$$J = J_{drift} + J_{diff} = 0$$

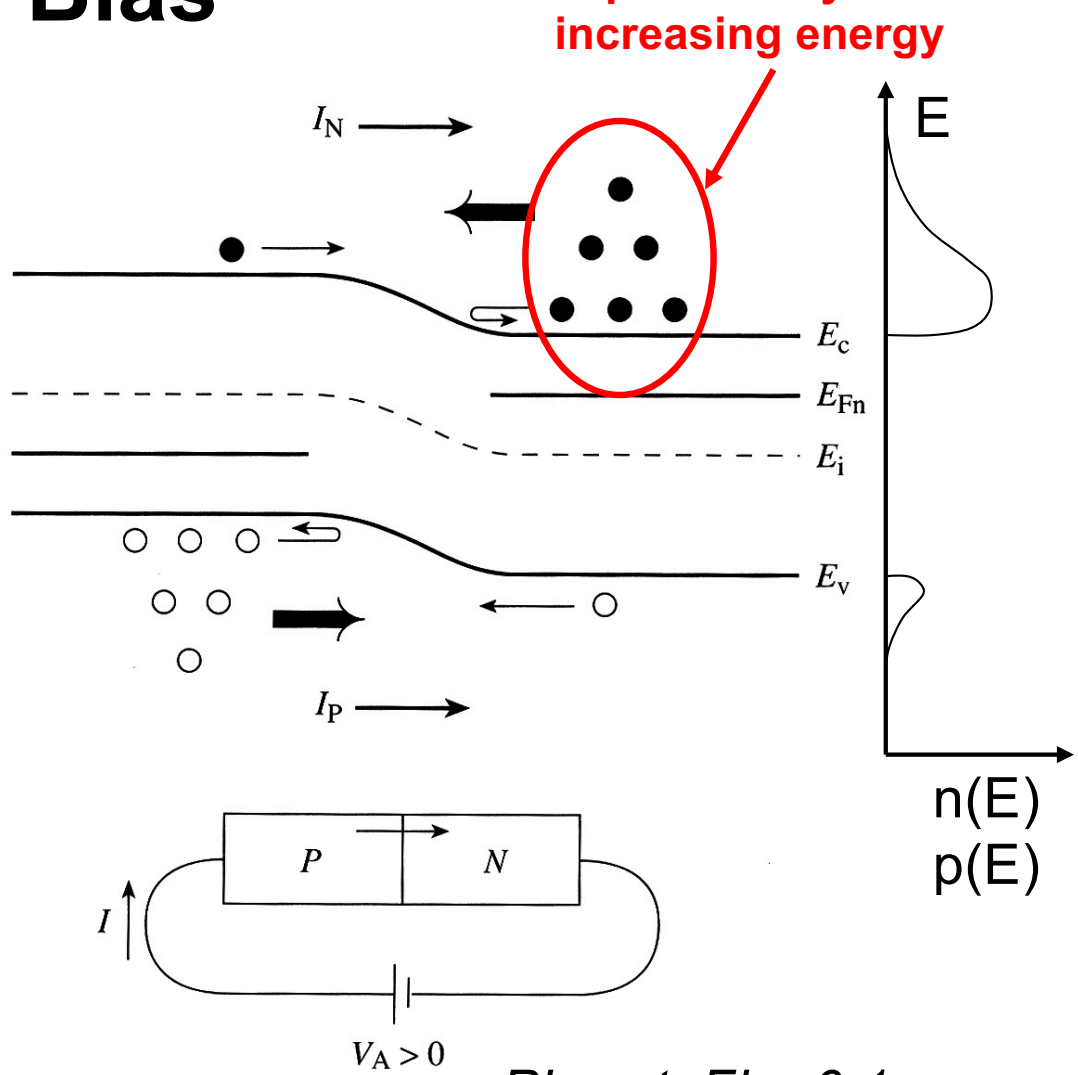


Pierret, Fig. 6.1

- Electron Currents:
  - Majority carrier diffusion current from n- to p-side
  - Minority carrier drift current from p- to n-side
- Hole Currents:
  - Majority carrier diffusion current from p- to n-side
  - Minority carrier drift current from n- to p-side

Qualitative Derivation (b) Forward Bias

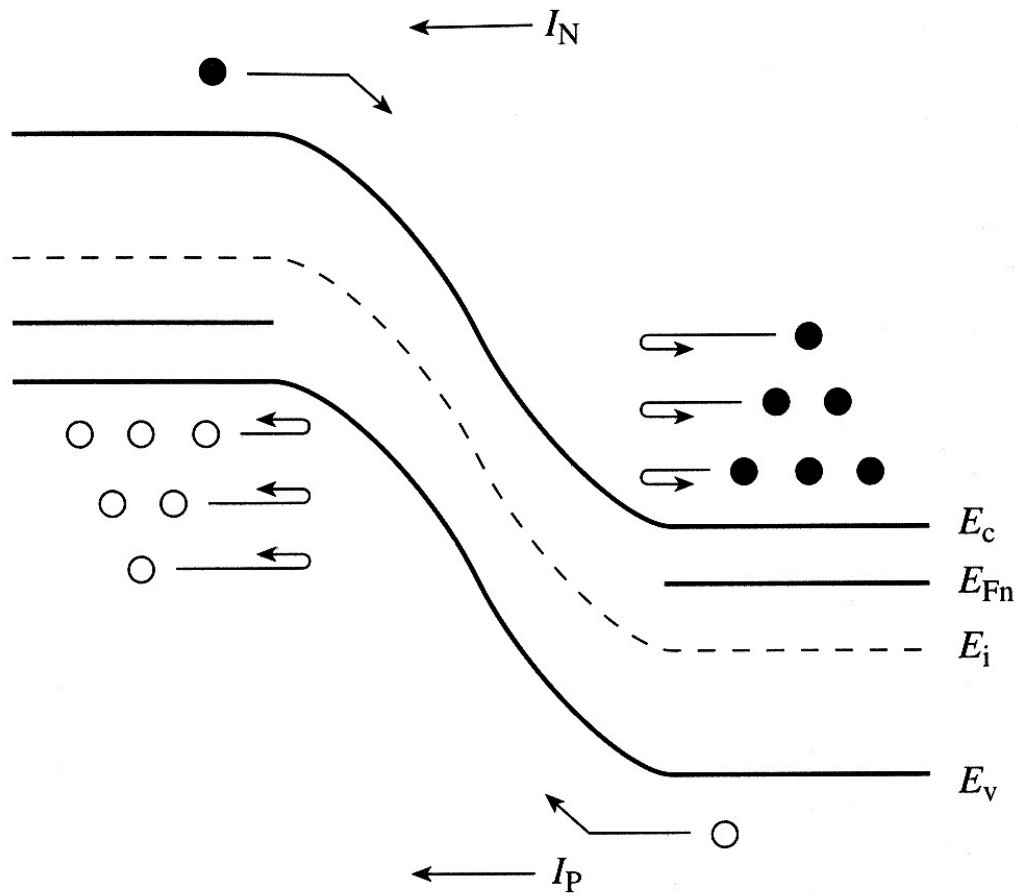
- Forward bias ($V_A > 0$) reduces the potential drop across the depletion region
- As a result, the majority carrier (electron and hole) diffusion over the potential hill increases exponentially, resulting in a net current flow across the junction
- The minority carrier drift currents remain constant



Pierret, Fig. 6.1

Qualitative Derivation
(c) Reverse Bias

- Reverse bias ($V_A < 0$) increases the potential drop across the depletion region
- As a result, the majority carrier (electron and hole) diffusion current over the potential hill is suppressed for $-V_A >$ few kT
- The minority carrier drift currents remain constant, thus resulting in a saturation current flowing in reverse direction



Pierret, Fig. 6.1

3.2.2 Quantitative Derivation – Solution Strategy –

1. **Calculate the minority carrier concentration** in the quasineutral regions on the p- and n-side of the junction by solving the continuity equation
2. **Calculate the minority carrier current densities** at the edges of the depletion region $x = x_n$ and $x = -x_p$
3. **Calculate the total current density** flowing through the diode by adding the hole and electron minority carrier density, assuming no generation/ recombination effects in the depletion region

1-D Continuity and Current Equations

- 1-D Current Density Equation

$$J_p = q\,\mu_p\,p\,\varepsilon - q\,D_p\,\frac{dp}{dx}$$

$$J_n = q\,\mu_n\,n\,\varepsilon + q\,D_n\,\frac{dn}{dx}$$

- 1-D Continuity Equation

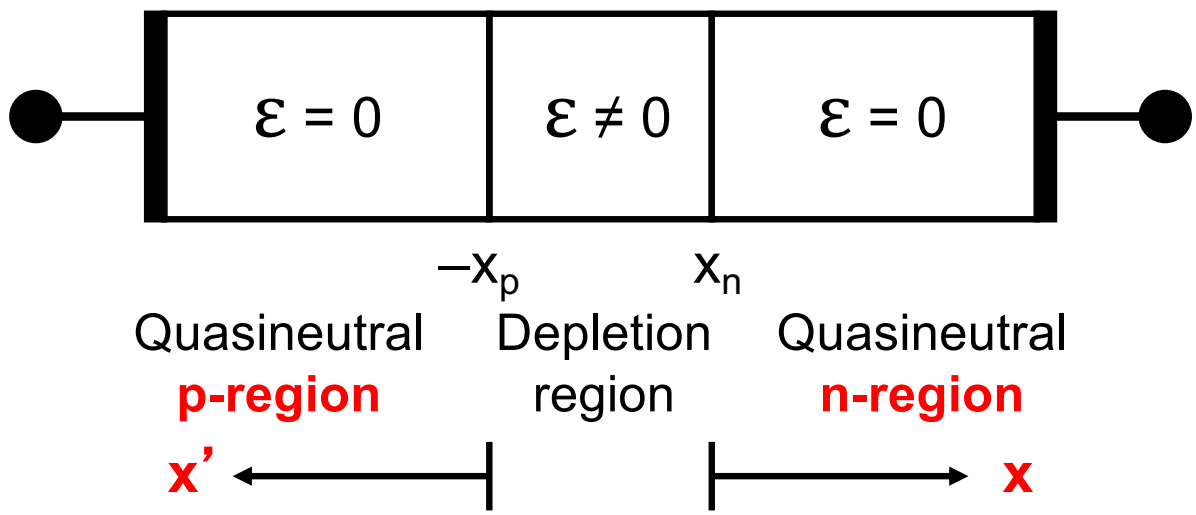
$$\frac{\partial n_p(x,t)}{\partial t} = +n_p\mu_n\frac{\partial \varepsilon}{\partial x} + \mu_n\varepsilon\frac{\partial n_p}{\partial x} + D_n\frac{\partial^2 n_p}{\partial x^2} + G_L - \frac{n_p - n_{p0}}{\tau_n}$$

$$\frac{\partial p_n(x,t)}{\partial t} = -p_n\mu_p\frac{\partial \varepsilon}{\partial x} - \mu_p\varepsilon\frac{\partial p_n}{\partial x} + D_p\frac{\partial^2 p_n}{\partial x^2} + G_L - \frac{p_n - p_{n0}}{\tau_p}$$

- Which terms do we have to consider?

PN Junction: IV Characteristic
Assumptions for Quantitative Derivation

- Steady state conditions
- Non-degenerately doped step junction
- One-dimensional pn-junction
- Low-level injection in quasineutral regions
- No generation/recombination processes other than thermal recombination/generation in the diode



PN Junction: IV Characteristic – Continuity Equation –

- Continuity equation for **quasineutral regions**
  - Steady state ($dp_n/dt = dn_p/dt = 0$)
  - No E-field, i.e. all drift terms are zero
  - No R-G other than thermal R-G (e.g., $G_L = 0$)

$$\boxed{\begin{aligned} 0 &= D_n \frac{d^2 \Delta n_p}{dx'^2} - \frac{\Delta n_p}{\tau_n} \\ 0 &= D_p \frac{d^2 \Delta p_n}{dx^2} - \frac{\Delta p_n}{\tau_p} \end{aligned}} \begin{aligned} &= D_n \frac{d^2 n_p}{dx'^2} - \frac{n_p - n_{p0}}{\tau_n} \quad \text{for } x' \geq 0 \\ &= D_p \frac{d^2 p_n}{dx^2} - \frac{p_n - p_{n0}}{\tau_p} \quad \text{for } x \geq 0 \end{aligned}$$

with $n_p = n_{p0} + \Delta n_p$ and $p_n = p_{n0} + \Delta p_n$

PN Junction: IV Characteristic
– Minority Carrier Concentration –

- General solution

$$\boxed{\begin{aligned}\Delta n_p(x') &= A e^{-x'/L_n} + B e^{x'/L_n} \\ \Delta p_n(x) &= C e^{-x/L_p} + D e^{x/L_p}\end{aligned}} \quad \begin{aligned}&\text{for } x' \geq 0 \\ &\text{for } x \geq 0\end{aligned}$$

with $L_n = \sqrt{D_n \tau_n}$ and $L_p = \sqrt{D_p \tau_p}$

- Boundary conditions at ohmic contacts:
  Assumption: wide quasineutral regions, i.e.

$$\Delta n_p\left(x' \to \infty\right) = 0 \quad \Rightarrow \quad \boxed{B = 0}$$

$$\Delta p_n\left(x \to \infty\right) = 0 \quad \Rightarrow \quad \boxed{D = 0}$$

PN Junction: IV Characteristic
– Minority Carrier Concentration –

- Boundary conditions at edge of depletion region:

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = \frac{kT}{q} \ln \frac{p_{p0} n_{n0}}{n_i^2} \overset{p_{p0}\, n_{p0} = n_i^2}{=} \frac{kT}{q} \ln \frac{n_{n0}}{n_{p0}}$$

$$\Rightarrow \boxed{n_{n0} = n_{p0}\, e^{qV_{bi}/kT}}$$

- Assumption: Equation is true also for applied V_A!

$$V_{bi} \rightarrow V_{bi} - V_A$$

$$\Rightarrow n_n = n_p\, e^{q\left(V_{bi} - V_A\right)/kT}$$

$$\Rightarrow n_p = n_n\, e^{-q\left(V_{bi} - V_A\right)/kT} \underset{n_n \approx n_{n0}}{\overset{\text{weak injection}}{\approx}} \underbrace{n_{n0}\, e^{-qV_{bi}/kT}}_{=n_{p0}} e^{qV_A/kT}$$

PN Junction: IV Characteristic
– Minority Carrier Concentration –

- Boundary conditions at edge of depletion region (cont.):

$$n_p = n_{p0}\, e^{qV_A/kT}$$

and similar

$$p_n = p_{n0}\, e^{qV_A/kT}$$

or

$$\Delta n_p = n_{p0}\left[e^{qV_A/kT} - 1\right] \quad \text{at } -x_p, \text{ i.e. } x' = 0$$

$$\Delta p_n = p_{n0}\left[e^{qV_A/kT} - 1\right] \quad \text{at } +x_n, \text{ i.e. } x = 0$$

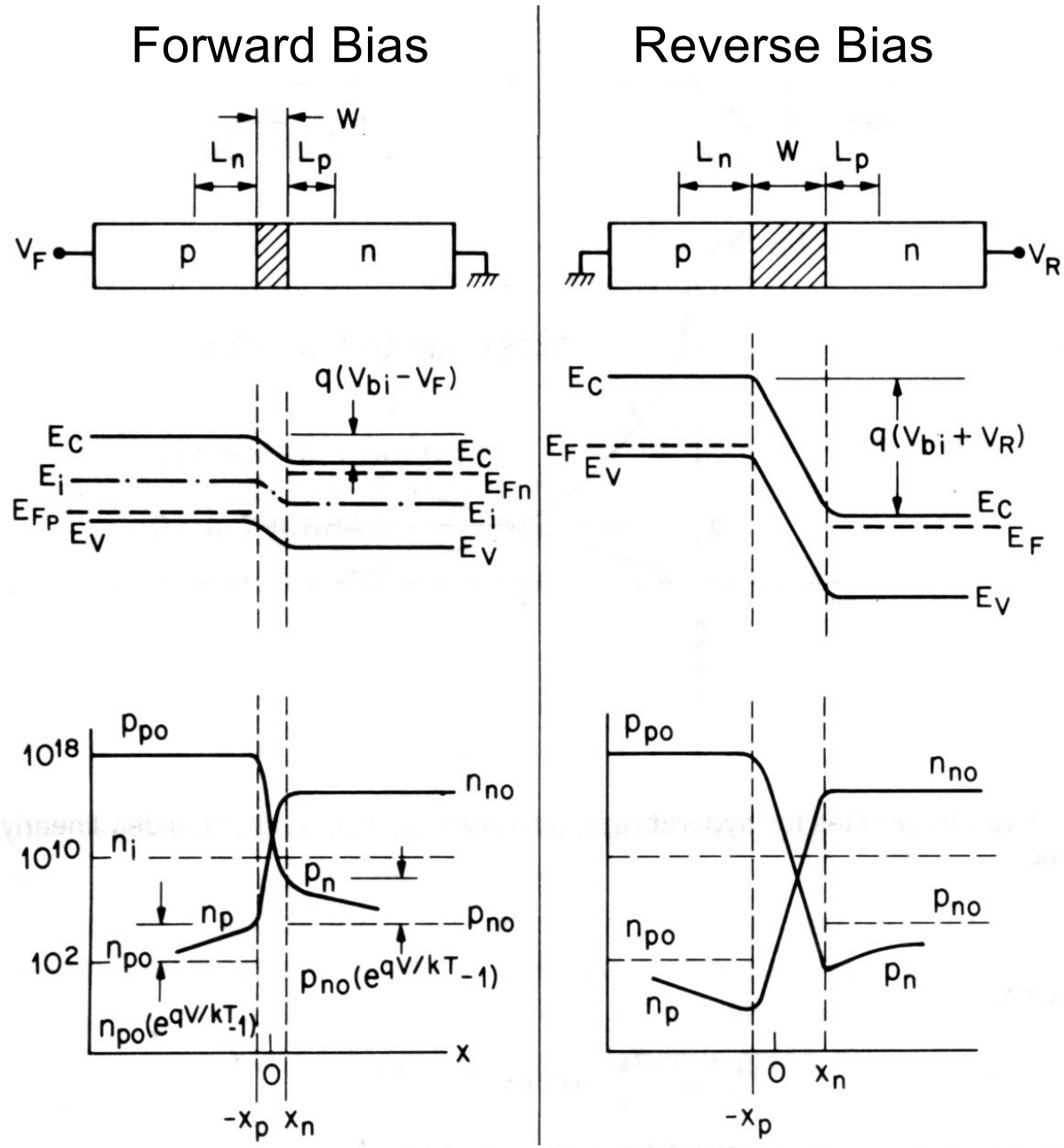
$$\Rightarrow \quad A = n_{p0}\left[e^{qV_A/kT} - 1\right] \quad \text{and} \quad C = p_{n0}\left[e^{qV_A/kT} - 1\right]$$

Minority Carrier Concentration

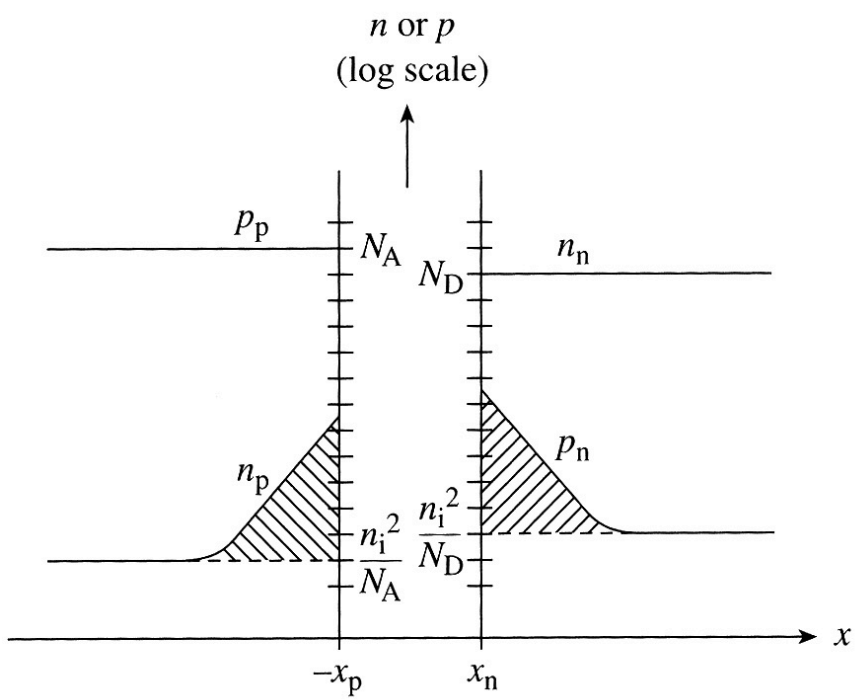
Thus, the minority carrier distributions in the quasi-neutral regions become:

$$\Delta n_p(x') = n_{p0}\left[e^{qV_A/kT} - 1\right]e^{-x'/L_n}$$

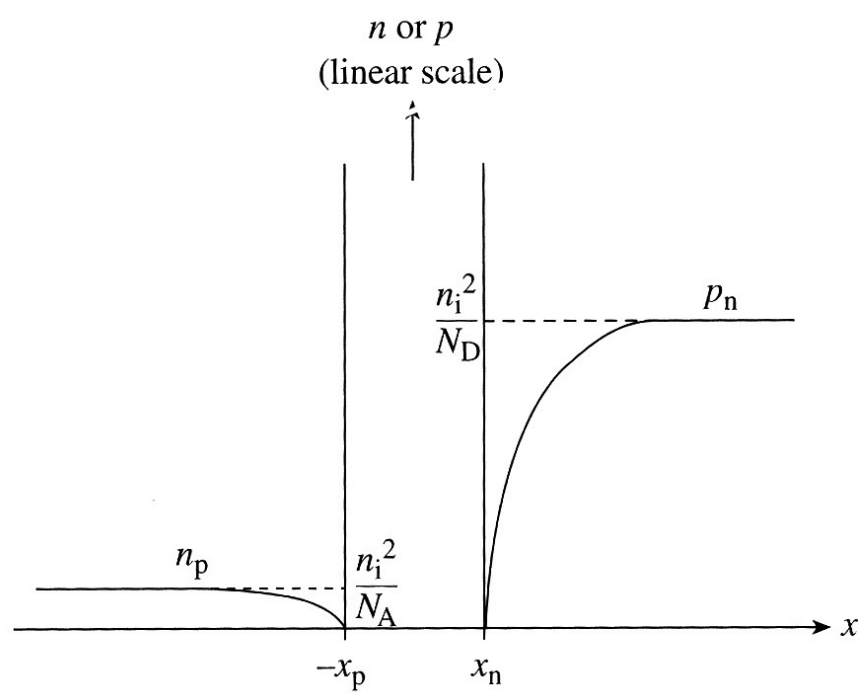
$$\Delta p_n(x) = p_{n0}\left[e^{qV_A/kT} - 1\right]e^{-x/L_p}$$



PN Junction: IV Characteristic
Minority Carrier Concentrations



Forward Bias



Reverse Bias

Pierret, Fig. 6.8

PN Junction: IV Characteristic
– Diffusion Current Densities –

- The minority carrier diffusion currents in the quasineutral regions can be derived from the minority carrier distributions by

$$J_n(x') = +qD_n \frac{d\Delta n_p}{dx'} = -\frac{qD_n n_{p0}}{L_n}\left[e^{qV_A/kT} - 1\right]e^{-x'/L_n}$$

$$J_p(x) = -qD_p \frac{d\Delta p_n}{dx} = +\frac{qD_p p_{n0}}{L_p}\left[e^{qV_A/kT} - 1\right]e^{-x/L_p}$$

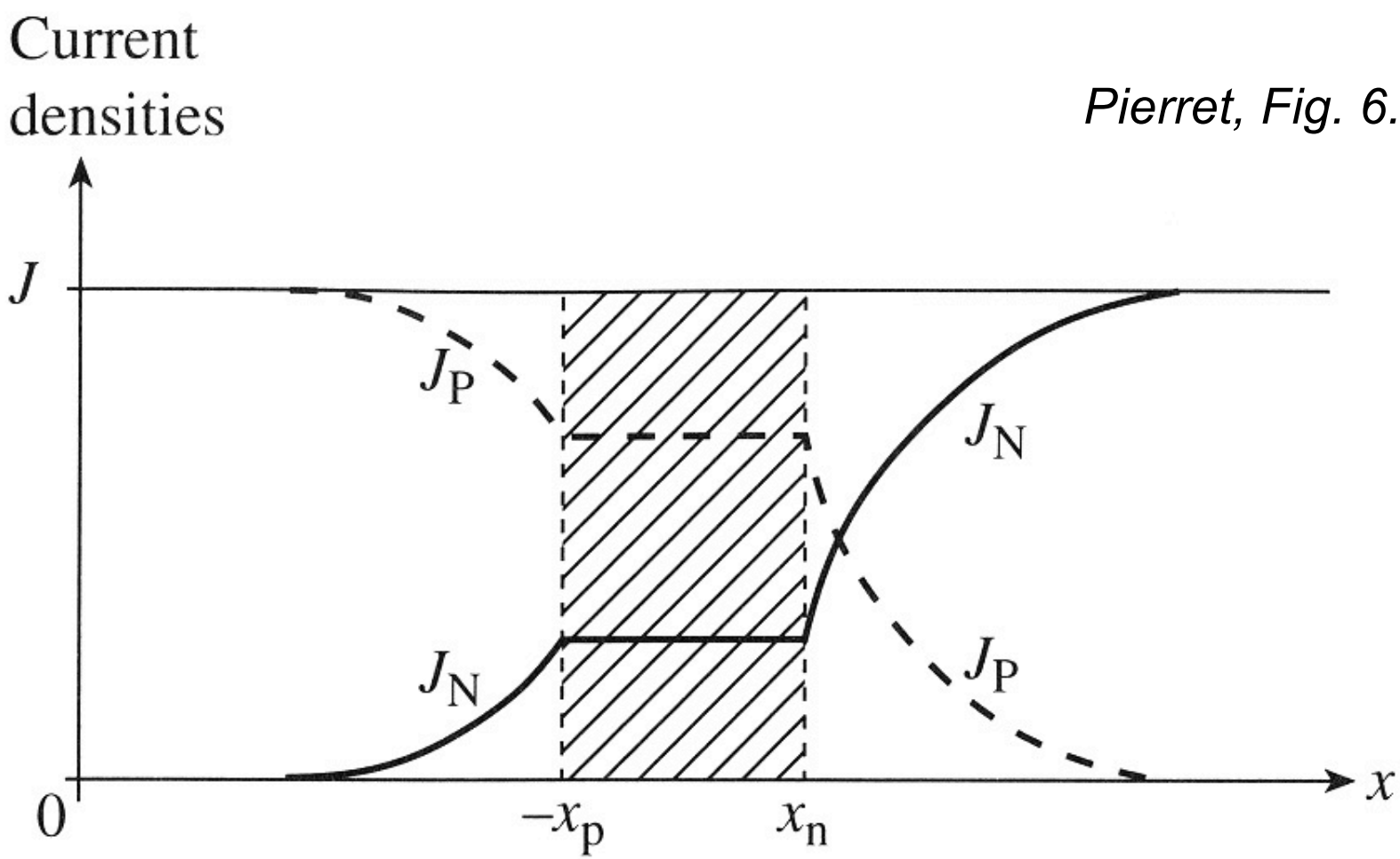
- Thus, at the edges of the depletion region

$$J_n\left(x = -x_p\right) = -J_n(x' = 0) = +\frac{qD_n n_{p0}}{L_n}\left[e^{qV_A/kT} - 1\right]$$

$$J_p\left(x = +x_n\right) = +J_p(x = 0) = +\frac{qD_p p_{n0}}{L_p}\left[e^{qV_A/kT} - 1\right]$$

PN Junction: IV Characteristic
– Diffusion Current Densities –

Pierret, Fig. 6.7



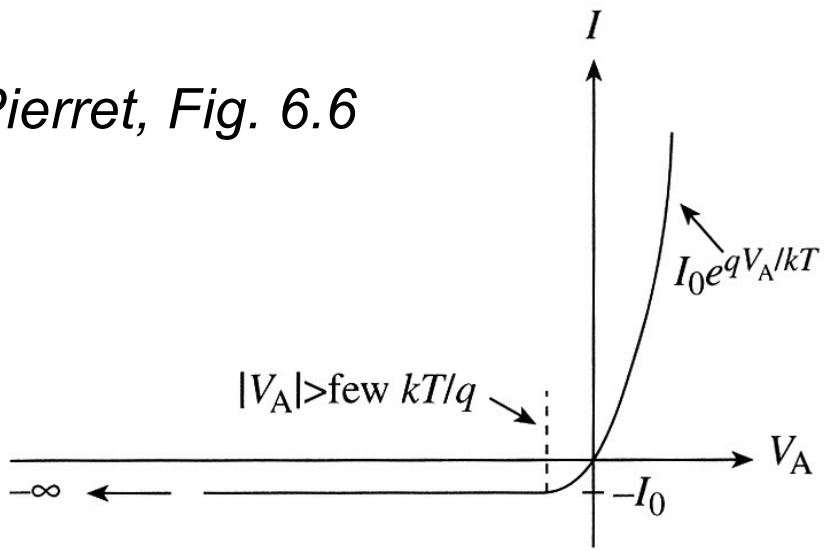
PN Junction: IV Characteristic
– Total Current –

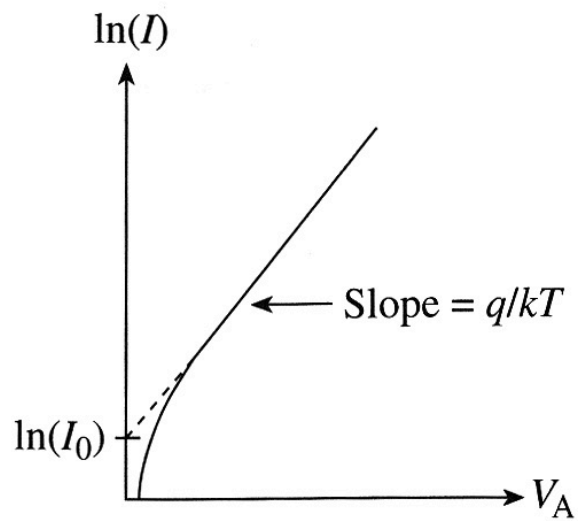
- Assuming that there is no R-G in the depletion region, the total current density flowing through the pn junction is the sum of the minority carrier currents at the edges of the depletion region

$$\boxed{\begin{aligned} J &= J_n\left(x = -x_p\right) + J_p\left(x = +x_n\right) \\ &= \underbrace{\left[\frac{qD_n n_{p0}}{L_n} + \frac{qD_p p_{n0}}{L_p}\right]}_{\equiv J_s}\left[e^{qV_A/kT} - 1\right] \end{aligned}}$$

PN Junction: IV Characteristic

Pierret, Fig. 6.6





Reverse Bias $V_A < 0$

- For $|V_A|$ > few kT/q, a small saturation current J_s is flowing through the pn junction

Forward Bias $V_A > 0$

- For $|V_A|$ > few kT/q, the current flowing through the junction increases exponentially

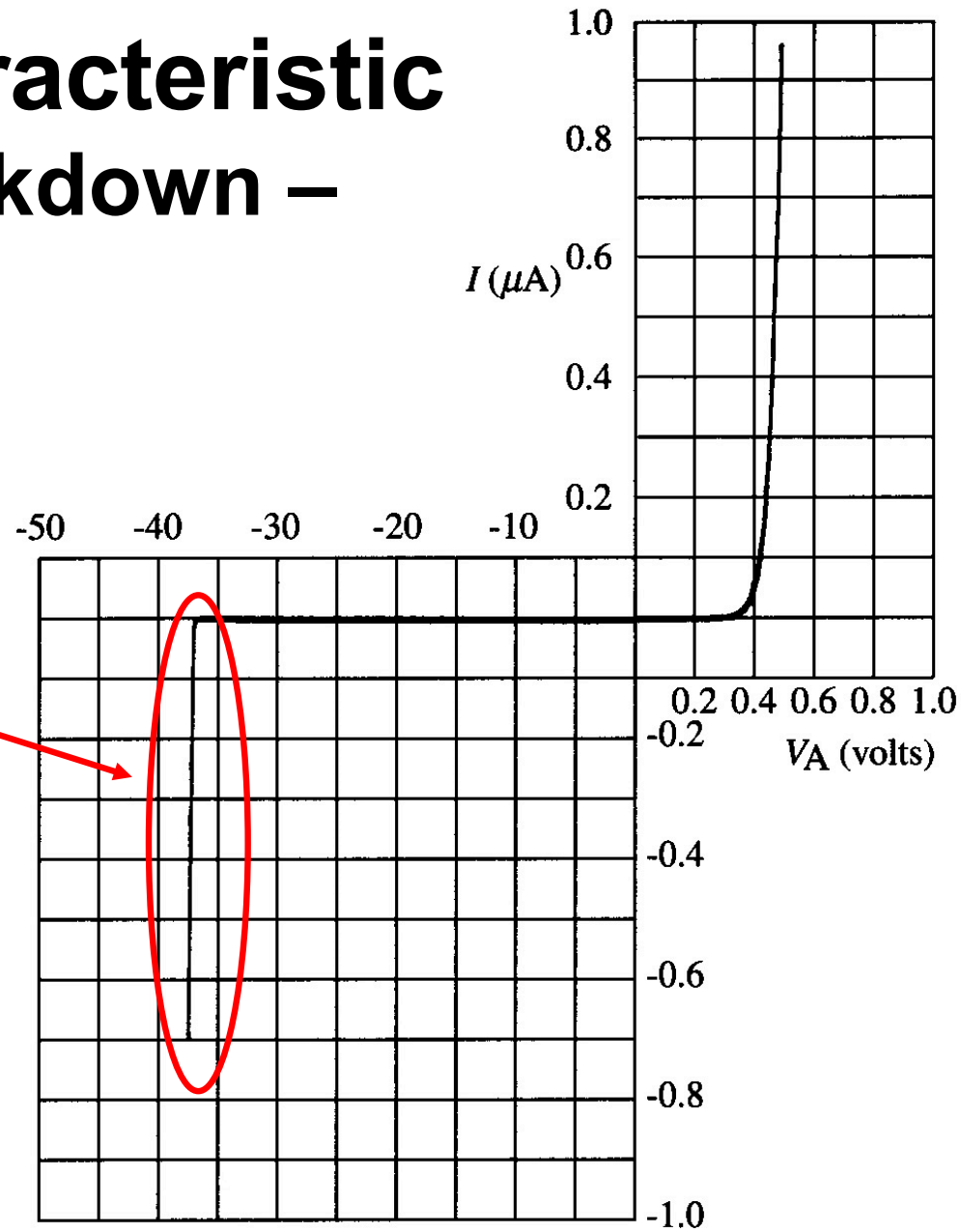
PN Junction: Saturation Current

The saturation current density is given by

$$J_s = \frac{qD_n n_{p0}}{L_n} + \frac{qD_p p_{n0}}{L_p} = \frac{qD_n n_i^2}{L_n N_A} + \frac{qD_p n_i^2}{L_p N_D}$$

- The saturation current depends on the **intrinsic carrier concentration n_i** and, thus, is strongly temperature and material dependent
- The saturation current is dominated by the contribution from the lower doped side of the junction in case of asymmetrical junctions

"Real" IV Characteristic
– Junction Breakdown –



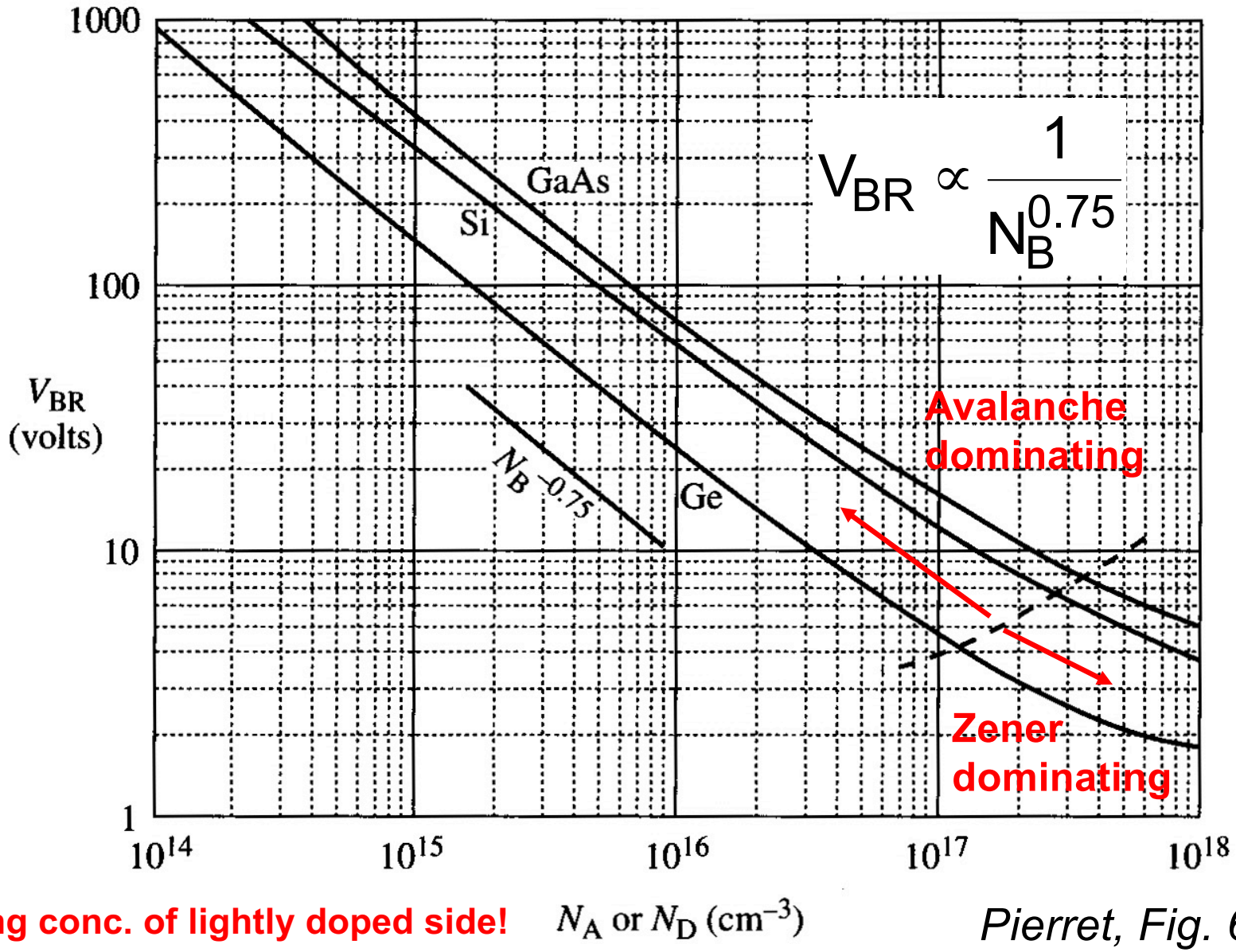
Junction Breakdown

Pierret, Fig. 6.9

3.2.3 Junction Breakdown

- **Junction or Reverse-Bias Breakdown:** current flowing in a pn-junction under reverse bias suddenly increases drastically if reverse bias is increased over the so-called **breakdown voltage V_{BR}**; breakdown process is reversible if the junction is not overheated (current must be limited)
- The **breakdown voltage** depends on
  - **Junction doping** (the non-degenerate doping in case of a n^+-p or p^+-n junction)
  - **Temperature**
  - **Radius of curvature** in case of junctions with curved non-planar edges because of larger electric fields in the curved edges; this effect becomes important for shallow junctions
- The basic **breakdown processes** are
  - **Avalanching**
  - **Zener process** (dominating if $V_{BR} < 4.5\ E_g/q$)

Breakdown Voltage in Planar p^+-n/n^+-p Junctions



Doping conc. of lightly doped side!

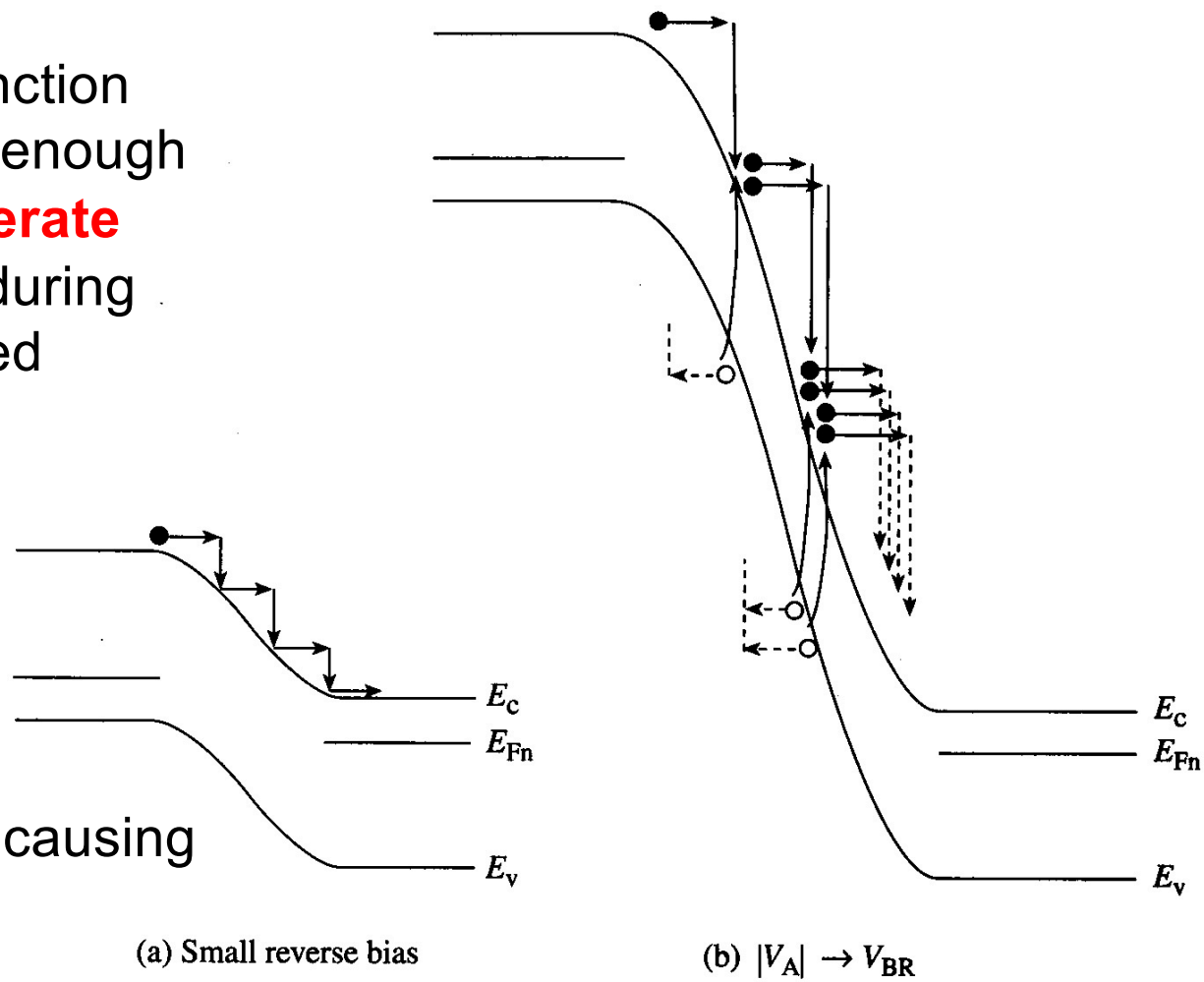
Pierret, Fig. 6.11

Avalanche Breakdown

Pierret, Fig. 6.12

Avalanching Process

- Minority carriers in junction depletion region gain enough kinetic energy to **generate electron-hole pairs** during an impact; this is called **impact ionization**
- This way, the number of carriers can increase tremendously, similar to a snow avalanche, ultimately causing a junction breakdown

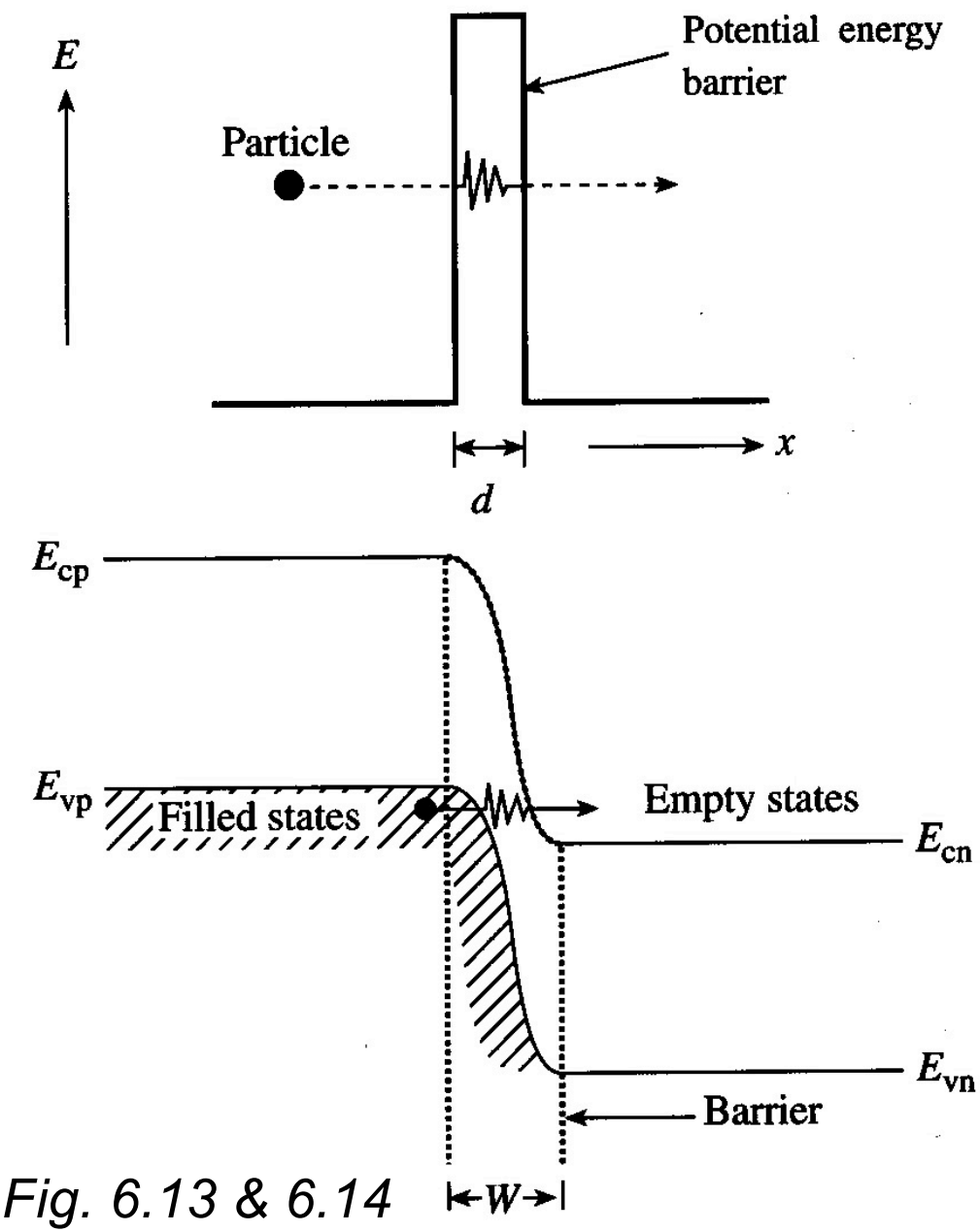


(a) Small reverse bias

(b) $|V_A| \rightarrow V_{BR}$

Zener Breakdown

- Breakdown current due to quantum mechanical **carrier tunneling** in highly-doped, degenerate pn-junctions
- Physical background: there is a quantum mechanical probability that a particle can tunnel through a potential barrier even if its energy is smaller than the barrier height



Pierret, Fig. 6.13 & 6.14